# MANISH HIRA

**Data Science & Analytics Leader | AI/ML Engineering & Strategic Innovation**

Richmond, VA · hiram@ieee.org · linkedin.com/in/mhira · Google Scholar

## Executive Summary

Senior Data Science and Analytics Leader with 20+ years of experience steering high-impact AI/ML engineering, predictive modeling, and enterprise data strategy across financial services (Capital One) and global technology firms (Samsung, Wipro). Ph.D. Candidate (VCU), M.S. in Computational Finance (UW), and MBA (Notre Dame). Proven track record of informing $50M+ marketing allocations, authoring 3 granted patents, leading cross-functional teams, and driving enterprise GenAI and agentic workflow adoption.

## Core Competencies & Technical Skills

- **Machine Learning & Statistics:** XGBoost, PyTorch, TensorFlow, Scikit-Learn, Sufficient Dimension Reduction (SDR), Causal Inference, Time Series Analysis (GARCH), A/B Testing, Feature Engineering, Bayesian Networks
- **Big Data & Cloud Architecture:** Snowflake, Databricks, AWS (Redshift, EMR, Glue, Lambda, SageMaker), PySpark, SQL, Python, R, C/C++, Unix Shell, ETL/ELT Pipelines
- **AI Innovation & GenAI Tooling:** Agentic Workflows (Claude Code, Gemini Gems), Copilot, LLaMA Sandbox Integration, Data Governance Automation, Model Audit & Validation
- **Leadership & Business Strategy:** Portfolio Management, Executive Communications, Agile/JIRA, Cross-Functional Team Leadership, Vendor Data Management, Product & Brand Intelligence

## Professional Experience

**Capital One**, Richmond, VA — January 2019 – Present
*Data Analysis Manager*

- Architected end-to-end vision for an enterprise **Agentic Brand Marketing Analytics Stack**, integrating multi-modal signal sources (1st-party transactional, market, competitive, and unstructured creative assets via OCR/vision tagging) into a unified 5D signal cube (*Channel* × *Touchpoint* × *Segment* × *Creative* × *Time*).
- Designed a hierarchical multi-agent orchestration framework (Competitive Intel, Media Mix Optimization, Audience Segmentation, Creative Intel) grounded in a governed enterprise semantic layer (SOV, SOW, NAB metrics) with strict Human-In-The-Loop (HITL) compliance guardrails.
- Built automated persona-based output pipelines delivering real-time tactical workbenches for channel managers, executive briefs for CMO/Board leadership, and decision-trace telemetry for econometrics/MMM teams.
- Evaluated multi-market advertising experiments across 10 major launch regions, directly optimizing $50M in media allocation, and authored 20-year media investment analysis establishing baseline brand marketing spend at 20% of total budget.
- Built predictive market opportunity model using XGBoost combined with Experian TAPS credit spending data; designed purchase volume (PVOL) geographic targeting analysis, compressing turnaround from weeks to days.
- Spearheaded enterprise AI tooling adoption across team: deployed custom Gemini Gems for standardized tech templates and data governance, and integrated GitHub Copilot and Claude Code for agentic workflow optimization.
- Led data governance and big-data engineering initiatives: migrated vendor data from spreadsheet silos to Snowflake and resolved processing bottlenecks on high-volume clickstream datasets.
- Managed a team of analysts and data engineers, overseeing talent development, performance management, and Program Increment (PI) technology roadmaps.

**State of Virginia / Capital One**, Richmond, VA — November 2015 – December 2018
*Data Analytics Consultant*

- Executed data science and analytics consulting engagements for financial services and retail enterprise clients.
- Designed targeted card marketing campaigns featuring customer segmentation for 50M+ consumers and $500M+ quarterly transaction volume.
- Developed XGBoost predictive models (AUC: 0.81) utilizing credit card transaction data to identify first-time retail partner shoppers.
- Engineered cloud-based ETL/ELT data pipelines from operational staging to enterprise data warehouse using Unix scripting and Oracle.

**Samsung Electronics**, South Korea & India — January 2009 – August 2013
*Senior Manager*

- Directed product portfolio planning for mobile division, directly impacting global sales of ∼100M smartphone units.
- Founded Intellectual Property (IP) Cell for Innovation Management, originating an institutional IP pipeline that yielded 18 granted patents.
- Built enterprise Business Intelligence (BI) infrastructure and dashboards utilizing MS SQL Server and SSIS/SSRS.
- Led and mentored an engineering team developing embedded software for mobile devices.

**Wipro Technologies**, USA, Finland, Japan, India — September 2000 – January 2009
*Technical Lead*

- Managed $10M+ multimedia software development for Qualcomm, resolving OEM commercialization blockers for MSM7x00 chipsets.
- Engineered automated mobile testing framework for Nokia Mobile (Finland) using Linux open-source software on TI OMAP platforms.
- Coordinated cross-border development of $2M DICOM-compliant MRI scanning software for Toshiba.

## Education

**Virginia Commonwealth University**, Richmond, VA — 2024 – Present
Ph.D. Candidate, Systems Modeling and Analysis

- Joint Statistical Sciences, Operations Research & Applied Mathematics program. Research focus: Sufficient Dimension Reduction (SDR) and causal inference in high-dimensional financial modeling.

**University of Washington**, Seattle, WA — 2019 – 2021
M.S., Computational Finance & Risk Management (GPA: 3.7)

- Mortgage Default Credit Risk Model using ML (Fannie Mae dataset; XGBoost, Random Forest, DNN).

**University of Notre Dame**, Mendoza College of Business, Notre Dame, IN — 2013 – 2015
MBA, Consulting & Finance (*Graduated Cum Laude*, GPA: 3.6 | GMAT: 750)

- *Arthur Foundation Fellow* (100% merit tuition scholarship).

**University of Delhi**, Netaji Subhas Institute of Technology (NSIT), Delhi, India — 1996 – 2000
B.E., Instrumentation & Control Engineering

## Granted Patents, Key Publications & Presentations

- **Granted US & International Patents:**
  - *Smart Card Authentication System*: US Patents 11,836,733 & 12,154,116 (Capital One Services, LLC).
  - *Near Field Communication (NFC) Method & System*: US Patent 9,160,418 & India Patent IN954DE2012 (Samsung Electronics).
- **Selected Publications & Conference Presentations:**
  - *Scientific Culture (2026)*: "Authenticity-Creativity Trade-offs in Generative AI Representations of Cultural Heritage: Voice AI Integration in Music Reconstruction" (Vol. 12, No. 5, pp. 2719–2733).
  - *MSI/ARF Analytics & Forecasting Conference (NYC, Sep 2026)*: Featured Presentation — "GEO in the Loop: Integrating Generative Engine Optimization into Marketing Analytics Workflows."
  - *All Day AI 2026 Virtual Conference (Oct 2026)*: Invited Speaker (BUILD Track) — "GEO in the Loop: A Framework for Integrating Generative Engine Optimization Into Marketing Analytics Workflows."
  - *ANA Analytics & Data Science Committee (Nov 2026)*: Invited Featured Speaker — "Gen AI Search in the Loop: Integrating GEO into Marketing Analytics Workflows."
  - *Temple Fox AIML Conference (Dec 2026)*: Accepted Paper — "Generative Engine Optimization in Enterprise Marketing: Multi-Stage Neural Dynamics & Causal Attribution."
  - *IEEE AISP Conference Proceedings (2026)*: "Integrating Deep Learning and Traditional Techniques for Image Enhancement."

## Executive Service, Industry Advisory & Honors

- **Executive & Industry Judging:** Judge, CODiE Awards (Best AI Data & Analytics Solution, 2025–2026) | Judge, AI Excellence Awards (2026) | Judge, RVAHack26 (ACM / IEEE, 2026) | Judge, CES Awards (2025) | Judge, AI for Good Hackathon (2025).
- **Committee & Peer Review Leadership:** Committee Member, Capital One Modeling & Analytics Conference (MAC 2026) Breakouts Committee | Featured Speaker, Richmond DA Connect (2026) | Peer Reviewer & Technical Program Committee for International & IEEE Conferences (AICCC, AIPE, ICon INDIA, ICMACC, ICEC, AIHC, EDIGE, NILES, ICNCCom, ICIIBMS, ICCSTE, SPML, iGET, AMLDS, TEMSMET, ICEECE).
- **Professional Affiliations & Pro-Bono Advisory:** Sigma Xi (Full Member) | IEEE Senior Member | ANA Committees (AI, Analytics) | Pi Mu Epsilon | Youth STEM Mentor (VJAS) | Taproot Pro-Bono Consultant.

## Professional Certifications & Specializations

- **Cloud & Big Data Certifications:** Databricks Academy AI/BI for Data Analysts | AWS Certified Machine Learning – Specialty | AWS Certified Data Analytics – Specialty.
- **Advanced AI & Quantitative Certifications:** Explainable AI (XAI) Specialization (Duke University) | PyTorch for Deep Learning (DeepLearning.AI) | MLOps Specialization (Coursera) | Quantum Computing & Programming (QBronze) | Machine Learning Engineer Nanodegree (Udacity).